Assessment of Low-Carbon Development Technologies in the Iron and Steel Industry

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清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

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Project Timeline

清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr
Research Proposal & preparations																		
Overview of China's iron & steel industry																		
Data requirement & database investigation																		
Technology assessment																		
Accounting models & technology report									1									
Investigation on breakthrough technology industry chain																		
High spatial resolution data collation																		
Model development																		
Data analysis & Scenario design																		
Final report																		2

- Milestones
  - 1. Assessment of Low-Carbon Development Technologies in the iron and steel industry: Model & Report
  - 2. Study on the Evolution of the Industrial Layout of the iron and steel industry: Model & Report

Overview of technology in I&S

- China's status quo: 90% BF-BOF and 10% EAF, almost no DRI

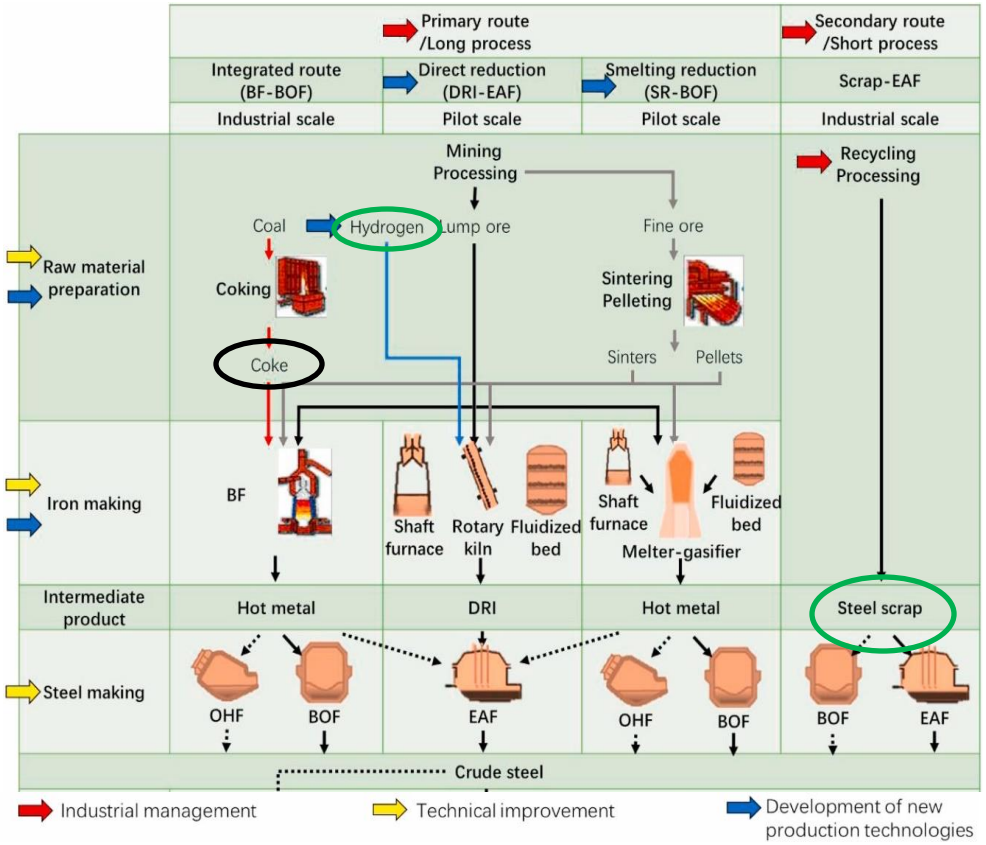
Technology	Description	Current Production Share (%)
BF-BOF	The Blast Furnace (BF) converts iron ore into molten iron using coke. The Basic Oxygen Furnace (BOF) then refines the molten iron into steel by blowing oxygen to reduce carbon content.	71%
EAF	The Electric Arc Furnace (EAF) melts scrap steel or Direct Reduced Iron (DRI) using electric arcs. EAF is highly flexible and energy-efficient.	24%
DRI-EAF	Direct Reduced Iron (DRI) is produced by reducing iron ore with natural gas or hydrogen. The DRI is then fed into an Electric Arc Furnace (EAF) for steelmaking.	5%

- Input stoichiometric data (ironmaking & steelmaking):

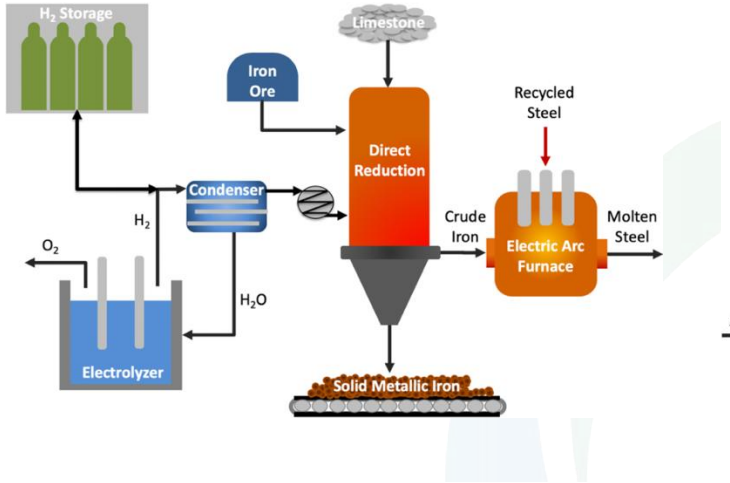
1. Material: iron ore, scrap, hot metal, DRI
2. Energy: coal, gas, hydrogen, electricity
3. Cost: CAPEX, OPEX, energy & material price
4. Emission factors

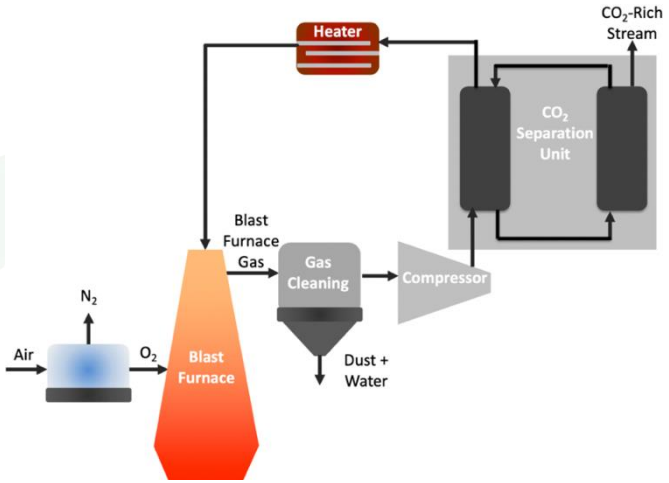
Typical production routes

- BF-BOF and scrap-EAF routes have established mature industry chains, for breakthrough technologies：
  - Green H_2: Renewable power generation, H_2 production from electrolysis, storage and transport, and consumption in steel plants.
  - Carbon Capture and Storage (CCS): CO_2 capture, transport, and storage

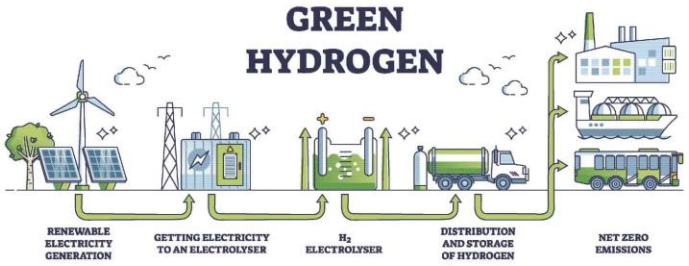


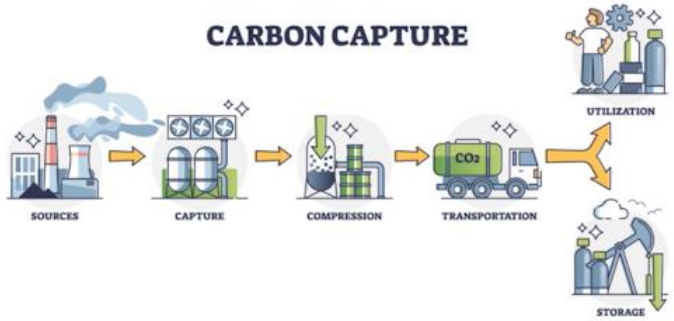
Typical crude steel production technical route





H_2-DRI

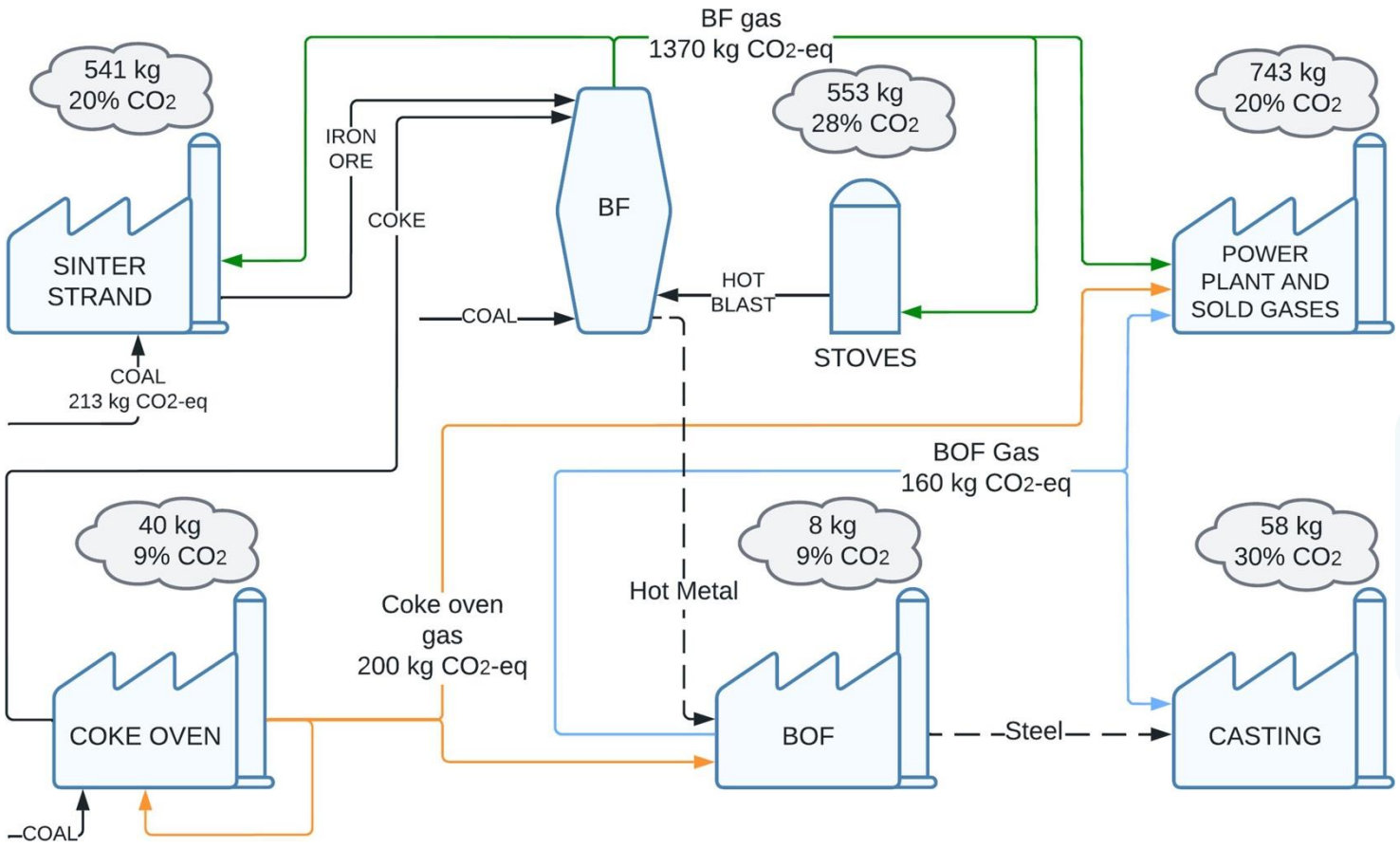


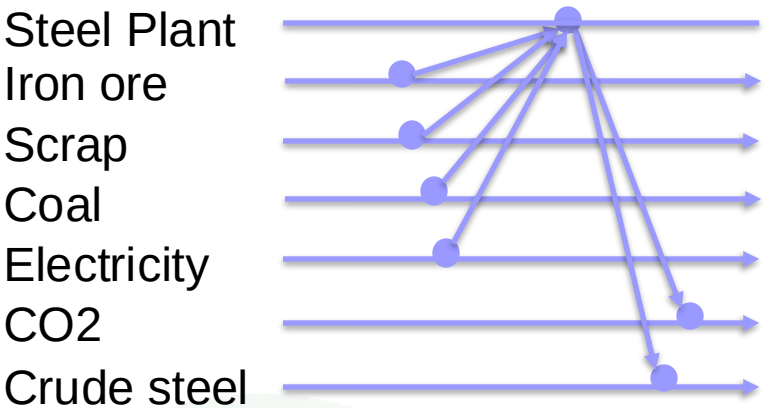


CCS

BF-BOF

- Strong integration of gas streams and heat in BF-BOF (low flexibility)
- Multiple emission sources, making CCS implementation difficult





Material consumption	t/t crude steel
Scrap	0.162
Iron ore	1.442
Hot metal	0.902

Source: China's Steel Statistical Yearbook (2022)
The input material mix for scrap-EAF is also obtained from yearbook

BF-BOF

- Scope: from raw material (iron ore & coal) to crude steel

Table 1
Process energy consumption intensity and fuel composition [37,72,76–83,93].

Process	Energy consumption intensity (MJ/t)	Fuel structure (%)								
		Coal	Oil	NG	Diesel	Gasoline	Fuel oil	Electricity	Coke	Coal gas
Underground mining	40.3	0	0	0	0	0	0	100	0	0
Opencast mining	13.8	0	0	0	77.1	0	0	22.9	0	0
Ore dressing	110.4	0	0	0	0	0	0	100	0	0
Sintering	1448.6	88.7	0	0	0	0	0	11.3	0	0
Pellet making	506.7	75.1	0	0	0	0	0	24.9	0	0
Scrap steel processing	93.9	0	0	0	7.9	0	0	92.1	0	0
BF ironmaking	14300.2	26.2	0	0	0	0	0	1.7	72.1	0
BOF steelmaking	255.3	0	0	0	0	0	0	100	0	0
EAF steelmaking	1333.2	0	0	0	0	0	0	100	0	0
Section bar processing	1866.4	0	0	0	0	0	0	21.6	0	78.4
Casting	258.3	0	0	0	0	0	1.4	45.1	0	53.5
Hot rolling	1716.2	0	0	0	0	0	0	74.8	0	25.2

Source:

1. Lei Ren et al., 2021, A review of CO2 emissions reduction technologies and low-carbon development in the iron and steel industry focusing on China, Renewable and Sustainable Energy Reviews

BF-BOF

Energy consumption	MWh/t crude steel
Coal	5.163
Electricity	0.198

Cost	/t crude steel
CAPEX	$ 600/t crude steel
OPEX	$ 18/y·t crude steel

Source:

1. Lei Ren et al., 2021, A review of CO2 emissions reduction technologies and low-carbon development in the iron and steel industry focusing on China, Renewable and Sustainable Energy Reviews
2. Actual project investment + experts interview
3. https://github.com/missionpossiblepartnership/mpp-steel-model

For BF-BOF-CCS:

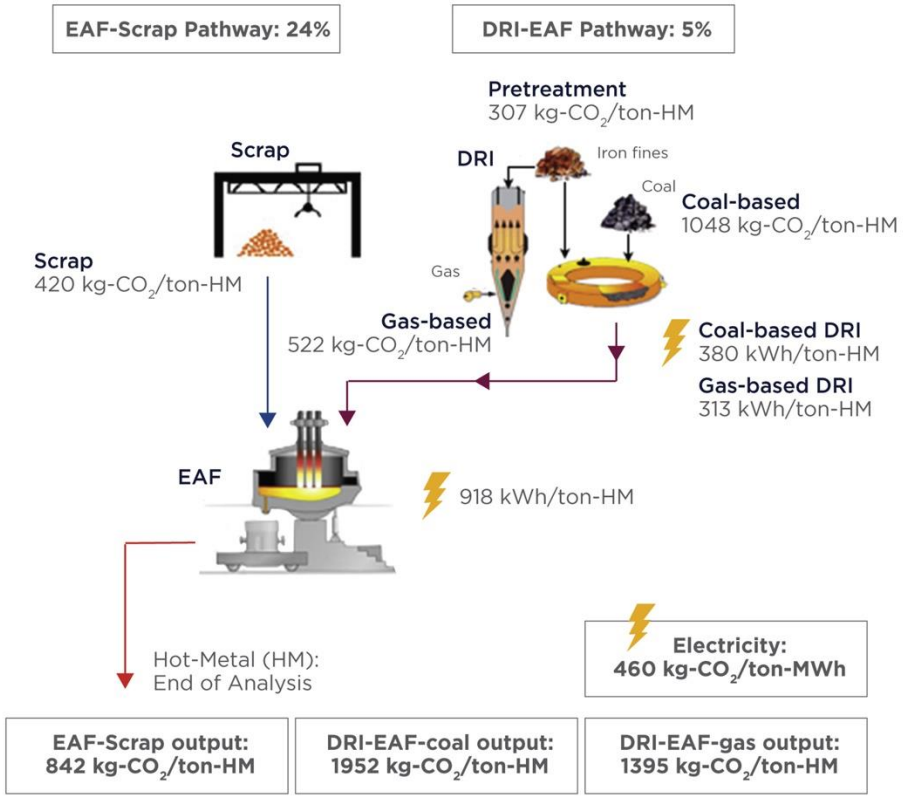
Energy consumption	MWh/t crude steel
Coal	7.010
Electricity	0.487

Cost	/t crude steel
CAPEX	$ 974.3/t crude steel
OPEX	$ 29.23/y·t crude steel

Source:

1. Kangkang Li et al., 2016, Systematic study of aqueous monoethanolamine (MEA)-based CO2 capture process: Techno-economic assessment of the MEA process and its improvements, Applied Energy (CCS CAPEX)
2. James, III, Robert E., Zoelle, Alexander, Keairns, Dale, Turner, Marc, Woods, Mark, & Kuehn, Norma. Cost and Performance Baseline for Fossil Energy Plants Volume 1: Bituminous Coal and Natural Gas to Electricity. United States. https://doi.org/10.2172/1569246 (CCS energy consumption)

Input data summary



Source: See table 2.

Note: EAF based design takes more electricity as energy input, provide additional opportunity for decarbonization.

Route	Energy type	Energy consumption	Unit	Source
BF-BOF	coal	5.163	MWh/t	Literature
BF-BOF-CCS	coal	7.010	MWh/t	Literature
scrap-EAF (~56%scrap)	coal	2.663	MWh/t	Yearbook
scrap-EAF (~100%scrap)	/	0.000	MWh/t	/
gas-DRI	gas	3.066	MWh/t DRI	Literature
gas-DRI-CCS	gas	3.881	MWh/t DRI	Literature
H2-DRI	H2	2.398	MWh/t DRI	Literature
DRI-EAF (~100%DRI)	/	0.000	MWh/t	/
gas-DRI-EAF	gas	3.429	MWh/t	/
gas-DRI-CCS-EAF	gas	4.341	MWh/t	/
H2-DRI-EAF	H2	2.682	MWh/t	/

Input data summary

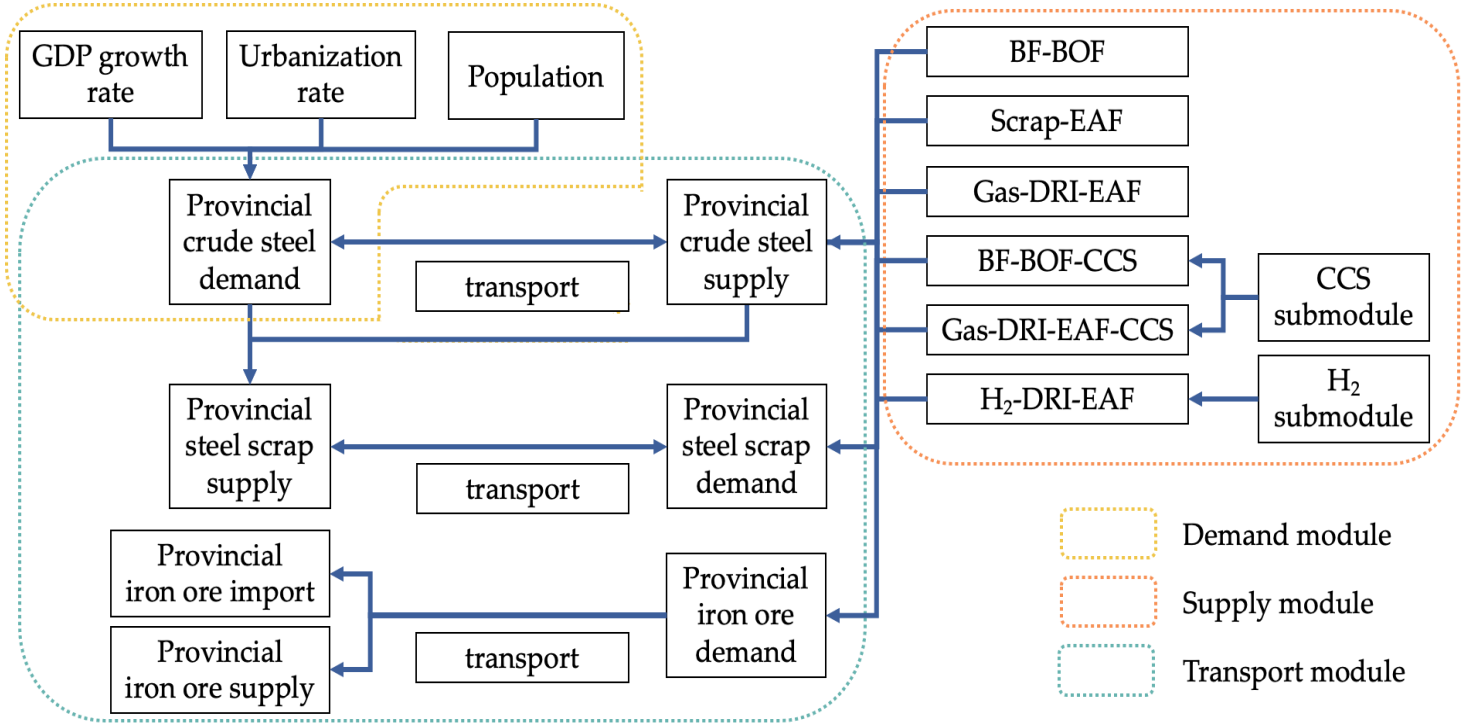
Route	Electricity consumption	Unit	Source
BF-BOF	0.198	MWh/t	Literature
BF-BOF-CCS	0.487	MWh/t	Literature
scrap-EAF (~56%scrap)	0.340	MWh/t	Yearbook
scrap-EAF (~100%scrap)	0.625	MWh/t	Interview
gas-DRI	0.162	MWh/t DRI	Literature
gas-DRI-CCS	0.289	MWh/t DRI	Literature
H2-DRI	0.285	MWh/t DRI	Literature
DRI-EAF (~100%DRI)	0.825	MWh/t	/
gas-DRI-EAF	1.006	MWh/t	/
gas-DRI-CCS-EAF	1.149	MWh/t	/
H2-DRI-EAF	1.143	MWh/t	/

Route	CAPEX	OPEX ratio	OPEX	Source
BF-BOF	600	0.141852	18.00	Actual project
BF-BOF-CCS	974.28	0.141852	29.23	Actual project + Literature
scrap-EAF (~56%scrap)	276.5	0.299934	8.30	Actual project
scrap-EAF (~100%scrap)	144.5	0.299934	4.34	Actual project
gas-DRI	350	0.169360	10.50	Literature + Actual project
gas-DRI-CCS	479.89	0.169360	14.40	Literature + Actual project
H2-DRI	350	0.169360	10.50	Literature + Actual project
DRI-EAF (~100%DRI)	276.5	0.299934	8.30	/
gas-DRI-EAF	626.51		18.80	/
gas-DRI-CCS-EAF	756.40		22.69	/
H2-DRI-EAF	626.51		18.80	/

Methodology

Multi-region planning model

- **Methodology: superstructure; Planning period: 2021~2060**
- **Temporal resolution: year; Spatial resolution: province**
- **Output: provincial pathway & industrial layout evolution**

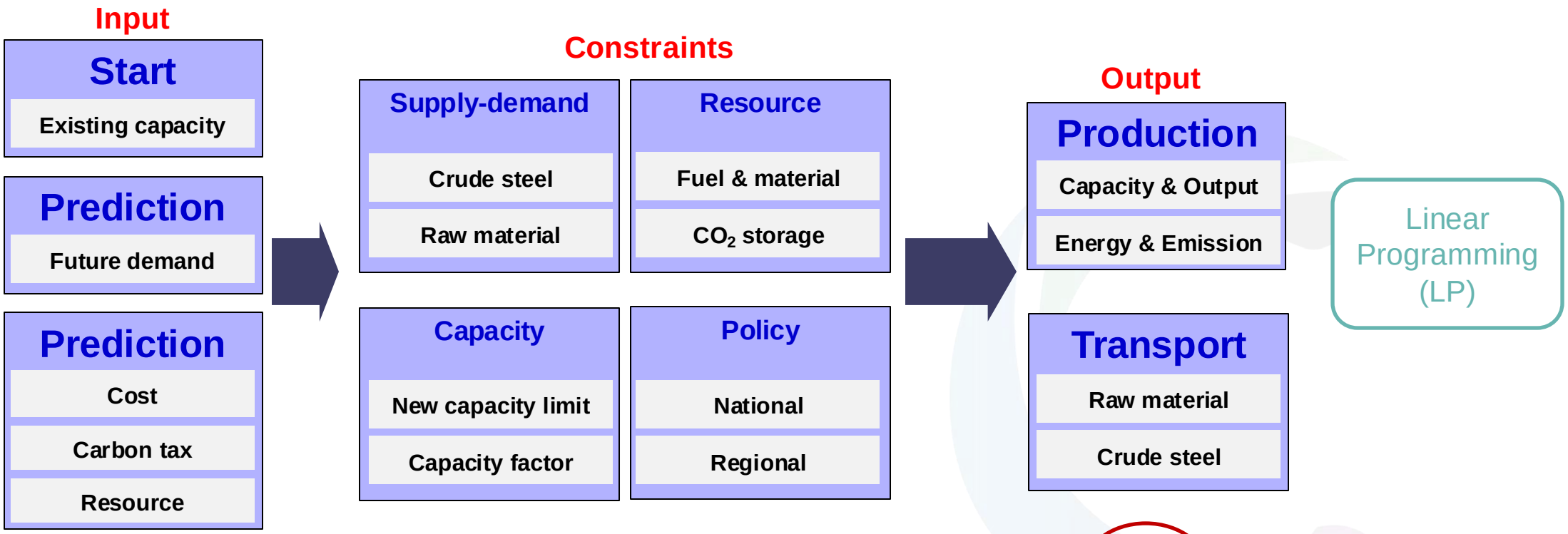


Model Output

Provincial energy mix	Provincial technology mix
Provincial CO_2 emission	Interprovincial transport
Provincial CO_2 storage	Production & transition cost

Multi-region planning model

- **Minimize the cost of supplying crude steel from an industry chain perspective**



$$\min cost = cost_{inv} + cost_{fuel} + cost_{ope} + cost_{mat} + cost_{tran} + cost_{emi}$$

total discounted cost

capital cost | fuel cost | O & M cost | Material cost | Transport cost | Emission cost

Driving factor: Emissions × Carbon Tax

Thank You For Your Attention!

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